

fit, transform

1. fit()

What it does:

Learns parameters from the data.

It does **not change the data**.

Example

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
scaler.fit(X_train)
```

What is learned?

- mean
- standard deviation

2. transform()

What it does:

Uses learned parameters to modify data.

Example

```
X_train_scaled = scaler.transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Important

- Never fit on test data.
- Only transform test data using training parameters.

3. fit_transform()

What it does:

Combination of `fit()` + `transform()` in one step.

Equivalent to:

```
scaler.fit(X_train)
X_train_scaled = scaler.transform(X_train)
```

Example

```
X_train_scaled = scaler.fit_transform(X_train)
```

Why useful

- Faster
- Cleaner code
- Used during training v